ESG Scores and Foreign Direct Investment (FDI): An Empirical Study of how ESG Scores Impact the Flow of FDI into Emerging Markets.

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Abstract

This thesis studies the relationship between ESG (environmental, social, and governance) factors and FDI (foreign direct investment) inflows and how this relationship affects FDI in developing markets. Panel data analysis is conducted to test the correlation between FDI inflows and sustainability factors, accounting for country-fixed effects for 6 select emerging countries within the East African region for the years between (2017-2022). I fit two main regression models for analysis, a pooled regression to cover all factors as one dataset, and a fixed-effects regression to account for country-specific variables. The explanatory analysis shall help host countries to understand and determine specific ESG factors that impact the growth of FDI inflows.

Keywords: *ESG (environmental, social, and governance), FDI (foreign direct investment)*

1. Introduction

Developments in investments have exhibited the incorporation of Environmental, Social, and Governance (ESG) factors as considerable measures of sustainability of investments in diverse markets. “The practice of ESG investing began in the 1960s as socially responsible investing (SRI), with investors building portfolios based on business activities” (MSCI, 2023). Within the financial investments domain, “sustainability looks at how an organization's products or services contribute to sustainable development” (Nordea, 2021).

The term ESG (Environmental, Social, and Governance) has been adopted in this domain to embody an investment approach that seeks to balance social benefits with financial returns. ESG investing has had an evolution from the traditional socially responsible investing paradigm. It offers a comprehensive framework for assessing social impacts beyond avoiding firms associated with negative outcomes (S&P Global, 2020).

International efforts including the United Nations' Sustainable Development Agenda and the Paris Agreement have established frameworks to advance this sustainable development over time. With the increasing significance of sustainability within economic disciplines, foreign investors must consider ESG factors alongside traditional investment models when evaluating potential investments in host or foreign countries. Substantial evidence of ESG factors significantly influencing Foreign Direct Investment (FDI) inflows, can help local authorities enhance ESG standards to attract more investment.

The growth and adaptation of ESG standards can not only improve the living conditions within host countries but also contribute to long-term value creation. Despite the increase in recognition of ESG factors in investment, existing research primarily focuses on two

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aspects: the impact of FDI on the economic development of the countries, and financial returns as a determinant of FDI location choices. Consequently, there remains a notable gap in quantitatively assessing the influence of ESG factors on FDI inflows for developing markets. The key focus will be on East African countries. Additionally, while some studies have examined the overall trends in FDI flows and ESG in emerging markets, they have not deeply explored how individual ESG components influence FDI at a regional level within Africa. This research seeks to bridge this gap by evaluating whether and how ESG considerations affect FDI inflows, thereby illuminating the significance of sustainability for foreign direct investors. Moreover, this study will identify which specific ESG factors are most significant to investors when committing capital to emerging and developing markets. By focusing on East Africa, this study not only contributes to the topic of sustainable investment but also provides actionable insights for policymakers and investors committed to fostering sustainable economic growth in the region.

Nature of ESG Factors

Evaluating the impact of ESG factors on Foreign Direct Investment (FDI) requires a good understanding of the varied nature of these factors. ESG factors can be categorized as discrete, continuous, categorical, constrained, ordered, or normalized. Each data type effectively presents unique challenges for analysis. Below I discuss briefly the overall nature of the respective factors.

Environmental (E) Factors

Environmental factors often have quantifiable metrics. These include carbon emissions (measured in parts per million), energy consumption (kilowatt-hours), water usage (liters), and waste management practices. These continuous variables present us with objective data that can be directly correlated with FDI inflows.

Social (S) Factors

On the contrary, social factors are more diverse and can include variables such as labor standards, community engagement, and human rights practices. They can be both quantitative (e.g., the number of local employees) and qualitative such as worker satisfaction, or categorical such as the presence of labor unions.

Governance (G) Factors

Governance factors look at corporate governance practices, regulatory environment, and political stability. They are often measured through subjective scores, such as corruption indices, and ease of doing business rankings. Governance factors may be discrete (e.g., the presence of anti-corruption laws) or normalized scores (e.g., governance quality index).

It is crucial to understand these differences for an accurate assessment of the impact of each ESG factor on FDI inflows. Due to the nature of the data, some factors may not be

directly quantifiable. I will therefore utilize composite indices provided by the World Bank data based on the combined ESG factors accordingly.

Goals and Objectives

This research aims to address the following critical challenges:

1. To determine whether there exists a relationship between Foreign Direct Investment (FDI) and the Environmental, Social, and Governance (ESG) factors of countries within developing markets.
2. In the case where we have such a correlation, which specific ESG factors highly influence FDI inflows, and what is the nature of their impact as drivers of FDI?

Research Questions and Hypotheses

The study aims to answer the following research questions:

1. How do environmental factors like renewable energy consumption influence FDI in East Africa?
2. What is the impact of social factors such as labor standards on FDI inflows?
3. How do governance factors, including political stability, affect FDI in the region?

Based on these questions, the following hypotheses are proposed:

H1: Environmental factors positively influence FDI inflows in East Africa.
H2: Social factors significantly impact the level of FDI in East African countries.
H3: Governance factors are critical determinants of FDI inflows in the region.

This research expects to observe a positive correlation between strong ESG factors and increased FDI inflows. Insights gained from this study will help policymakers measure and understand how foreign direct investors make decisions based on sustainability practices, and whether they perceive poor ESG ratings as a potential risk. They can then better determine enhancements that are essential for attracting greater levels of foreign direct investment.

2. Literature Review

Introduction

Over time, “the rise of ESG factors in the business world encourages investors from various countries to emphasize and pay attention to responsible investment business practices now as a feature of sustainable investment” (Yu et al., 2018). This section explores literature review on the key driving factors of Foreign Direct Investment. Specifically, it narrows down to the theoretical background of FDI and presents empirical evidence concerning the effects of ESG on FDI.

Theoretical Background of FDI

Theory on FDI has evolved significantly over time. The concept was initially tied to classical economics, where differences in return rates across different countries would be used to quantify capital movements. Kindleberg (1969), posits that “in the presence of perfect competition, the only way to be part of the international market would be through international trade. In this scenario, market failures in a host country attract FDI, giving foreign institutions an upper hand when operating in such markets.” Progressive research has however seen modern theories expand to include more nuanced explanations of the existence of FDI.

The Eclectic Paradigm Theory

One of the most prominent frameworks of FDI is Dunning's Eclectic Paradigm, otherwise known as the OLI model (Dunning, 1988). This theory posits that FDI is driven by three primary factors that firms consider before operations in foreign markets: Ownership, Location, and Internalization advantages.

Ownership advantages entail intangible assets that are exclusive to a given firm. These include firm-specific assets such as owned technology or brand reputation. These properties can be transferred to subsidiaries in a foreign country at little to no cost, offering an advantage relative to competitors. Such advantages may also arise from factors such as economies of scale in the firm.

Location advantages involve country-specific factors such as natural resources or labor costs. These are economic advantages tied to a specific location originating from the cost of production, market size, and transportation. Cultural similarities and political stability are other examples of social advantages that fall under this category.

Internalization advantages are the benefits firms gain from controlling production rather than licensing or outsourcing. Whenever international market challenges make international operations costly and costs can be reduced by handling tasks internally,

then the firm is likely to invest abroad on its own, instead of issuing licenses or franchises to third parties.

Another key theory is the Product Life Cycle model proposed by Vernon (1966). This model suggests that FDI patterns follow the life cycle of a product, with firms initially exporting, then moving production abroad as products mature and cost pressures increase.

Internalization Theory

According to Hymer (1966), local firms in host countries exhibit better information on the local environment, unlike foreign investors. This points to an existing information asymmetry that challenges FDI growth. For FDI to grow under these circumstances, foreign firms must exhibit certain characteristics that facilitate such an investment. A key consideration is that there exists a demand for, and these advantages are only accessible to foreign investors. Buckley and Casson (1976) built further onto this idea by positing that "companies organize their internal activities to develop firm-specific advantages. As a result, instead of conducting business externally between two or more firms in different countries, a firm would rather set up a business unit in a foreign country and maximize profits by working internally across national borders." Hymer (1966) posits that such a firm "would only undertake said foreign investment if the benefits from exploiting the firm-specific advantages outweigh the cost of operating internationally."

Empirical Evidence on FDI Determinants

This section looks at firm-exogenous factors that control the flow of FDI, specifically institutional and macroeconomic factors. Macroeconomic Factors

Macroeconomic Factors

One of the key determinant factors of FDI inflows into developing countries is economic growth. Empirical evidence by De Vita et al. (2008) proves that host country GDP growth, used as an indicator of its economic state, has a positive correlation to FDI inflows. Dollery et. al (2010) posit that FDI inflows have a positive impact on economic growth with the existence of a skilled labor force. Several other researchers have also studied the correlation between FDI and GDP growth. Choe (2003) finds that the relationship between FDI and GDP is bi-directional, such that FDI causes GDP growth and vice-versa. Choe's study however reveals that the effect of GDP growth on FDI inflows is more prevalent than the effect of FDI inflows on GDP growth. Fedderke and Romm (2006) found that market size, usually measured by GDP, has a strong positive impact on FDI inflows in the South African market.

Empirical studies have also found that factors such as interest rate, exchange rate and its volatility, and trade openness have an impact on FDI inflows. In their study, Fedderke and Romm also prove that increased exports have a positive relationship with FDI while increased imports have a negative relationship with FDI, owing to a country's trade openness. As Asiedu (2002) presents, "trade openness promotes FDI inflows into Sub-Saharan and non-Sub-Saharan African countries". "In Australia, we observe similar results where trade openness is determined as a significant factor in attracting FDI" (Yang et al, 2000). "The exact impact of trade openness is however dependent on the type of FDI" (Aizenman and Noy, 2006). Vertical FDI which is more evident between developing and developed countries tends to create trade whereas Horizontal FDI, more prevalent in developed countries is trade substituting.

Sayek (2009) posits that "inflation rate, which is a measure of economic stability, is a key macroeconomic factor for attracting FDI inflow." The study revealed that inflation rate and FDI are significantly but negatively correlated. A possible cause for this is attributed to high inflation rates which reduce existing and prospective FDI flows via reduction of the value of real return expected from the investments. On the contrary, a positive relationship between FDI and 5 inflation is presented in Mason and Vracheva's (2017) research. This positive relationship, as Agudze and Ibhagui (2021) posit, could be due to an increase in price levels railing up the economy presenting investors with adequate returns on their investment.

When assessing exchange rates, currency depreciation in the host country often increases FDI inflows as it gets more affordable for foreign investors to acquire assets abroad (Froot and Stein,1991). From their investigation of the impact of exchange rate uncertainty on FDI, Goldberg and Kolstad (1994) identify that forex uncertainty increases FDI by risk-averse enterprises if the uncertainty possesses a positive correlation with export demand shocks in their target markets. However, Campa (1993) looked at how exchange rate uncertainty affects FDI based on option theory and found that greater uncertainty increases the value of a firm's option to wait until investing in a market, consequently reducing FDI.

For developed markets, FDI targets policy-related determinants such trade openness, freedom index, and GDP growth. On the contrary, in developing countries, FDI shows a positive association with efficiency variables, gross fixed capital formation (GFCF), and trade openness (Saini and Singhania,2008). Infrastructure development has also been identified as an important factor in attracting FDI. Kang and Lee (2007) posit that investors prefer to invest in a country with established infrastructure. One would therefore expect infrastructure development to have a significant positive relationship with FDI (Mina, 2007).

Existing literature further studies the impact of financial development on FDI flows. As presented by Alfaro et al. (2010), financial development is a contributory factor of FDI inflows. They also that FDI brings higher economic growth in countries

that have developed financial sectors, contrary to those with underdeveloped financial sectors. Further analysis by Rajan & Zingales (1998) also finds that functional financial markets in host countries reduce the cost of external finance for firms, consequently increasing FDI inflows, especially from firms that tend to be dependent on outside capital. Empirical research by Bilir et al. (2019) posits that financial development in a country is driven by relatively more entry by multinational firms in the host countries.

A notable point from the study is that higher financial development results in increased competition locally, which in turn reduces the profits of firms in the country. This could present a hindering factor on FDI if the multinational firm's primary market is the local market. Another crucial consideration is human capital. Research conducted by Chidlow et al (2009), shows that productivity as well as cost of labor have a significant correlation with FDI.

Institutional factors

Romm and Federrke, (2006) posit that "Institutional factors such as property rights and political risk are key considerations for foreign investors." As further shown by Dupasquier and Osakwe (2006), poor governance and unfavorable regulatory environments minimize FDI inflows.

Empirical Evidence on the link between ESG and FDI

Existing literature on FDI and ESG has mostly looked at the relationship between FDI and specific ESG factors exclusively. There exists a gap in literature focused on the relationship between FDI and all three factors combined. Studies have shown that African countries attract FDI due to their abundant natural resources and growing consumer markets. For example, Asiedu (2002) found that natural resource endowments and market size significantly influence FDI inflows to Africa. However, institutional quality, including political stability and regulatory frameworks, also plays a crucial role (Busse & Hefeker, 2007). The following literature explores the above factors concerning FDI.

Relationship between FDI and Environmental Factors

The relationship between FDI and environmental factors is pegged on two main hypotheses: The pollution halo hypothesis and the pollution haven hypothesis.

The Pollution Halo Hypothesis

The pollution halo hypothesis posits that "Foreign Direct Investments help in reducing pollution via technology transfer and adoption of best practices to host countries from other countries" (Gallagher et al., 2007). In their study, Pisani et. al

(2019) found that "greener" cities which they considered to have good air quality and proper wastewater treatment attract FDI flows. They also note that multinational companies prefer to invest in these cities due to their commitment to the well-being of their employees and to improve their corporate image in the wake of awareness from stakeholders. Mert and Caglar (2020) look at the relationship between environmental pollution and FDI in Turkey.

Mert and Caglar (2020) in their reserch posit that increasing FDI will lead to a decrease in the rate of emissions growth in both the short and the long run, which they attribute to the transfer of technology, a more efficient approach to resources. Looking at the inverse relationship, Nadeem et. al (2020) find no conclusive result to back the hypothesis that FDI inflow leads to environmental degradation in Pakistan. A separate study by Xie et al. (2023) however finds that whereas FDI can positively moderate the impact of economic growth on environmental degradation by better technology, it ultimately increases industrialization which may regrettably increase environmental degradation.

Pollution Haven Hypothesis

The pollution haven hypothesis states that lower environmental standards attract FDI inflows (Copeland et al., 1994). Cole (2004) studied the impact of environmental regulation on FDI and observed that stringent regulations lead to reduced FDI. Dam and Scholtens (2012) add to this finding by observing that "multinational enterprises relocate their pollution-intensive operations to countries with more lenient environmental regulations". Spatareanu (2007) finds that FDI in a host country is correlated with strict environmental regulations in the investor's home country. On the other hand, Bu and Wagner (2016) posit that firms' heterogeneities in environmental capabilities and sizes have an impact on their investment patterns. This simply means that firms with higher environmental capabilities can invest in countries with stricter regulations whereas those with lower environmental capabilities tend to naturally shy away from investing in markets with strict environmental regulation.

Relationship between FDI and Social Factors

Benhabib et al., (1994) posit that a greater level of human capital is a derivative of higher life expectancy, higher economic living standards, as well as high education standards. According to Sharma and Gani (2004), their analysis of the Human Development Index (HDI) presents that FDI has a positive effect on human development. According to Li and Lu (2005), the level of human capital is a key contributor to FDI inflows. Kheng, Sun, and Anwar (2017) evaluate the correlation between human capital and FDI and they find significant causality between FDI and human capital.

Relationship between FDI and Governance Factors

Good governance, responsible institutions, and lower levels of corruption attract FDI inflows (Buchanan et al. 2012). Bailey (2018), further confirms this in his finding that democracy, political stability, and the rule of law encourage FDI, while factors such as corruption, heavy taxation, and cultural differences discourage FDI. Globerman and Shapiro (2002) define governance infrastructure as "an effective, impartial and transparent legal system that protects property and individual rights; public institutions that are stable, credible and honest; and government policies that favour free and open markets". In their study of the relationship between governance infrastructure and FDI inflows and outflows, they find that investing in governance infrastructure is a crucial determinant for FDI inflows. Good governance infrastructure creates an environment where domestic firms can grow and invest internationally. "The benefits of investing in governance infrastructure to promote FDI inflows are more pronounced in smaller and less developed economies" (Globerman et al., 2002).

Competitor Analysis

Empirical research on the relationship between ESG factors and FDI has gained value with the deepening of economies of trade. The systematic and comprehensive study of the correlation between these variables has been crucial since the 20th century, back in the 1980s. Academic literature on this correlation has been abundant in recent years, but obtaining a clear conclusion has been difficult. There is also a lack of sufficient evidence on the impact of ESG on East African countries' FDI, both positive and negative. Sufficient information and understanding of the correlation discussed above would also contribute to developing an ESG market index for the East African markets. This, I believe is a key gap that this research aims to fill.

Wang, et al. (2021) investigated the relationships between FDI and select ESG factors in six African nations for the period between 1971 and 2009, including Kenya, Zambia, and South Africa. Applying the Granger causality test for analysis, they discovered that "FDI has a significant positive correlation between CO2 emissions in Zimbabwe and a negative and significant effect on emissions in the Democratic Republic of the Congo, South Africa, and Kenya, but an insignificant effect on emissions in all other countries." This points to an inconclusive result when looking at environmental factors concerning FDI.

Wanjiru and Mwangi (2018) explored the impact of sustainability practices on FDI inflows in East Africa. Their study evaluated the positive correlation between environmental regulations and FDI, positing that investors increasingly favour countries with strong environmental policies. However, 8 their research excludes social and governance factors which also have an impact on FDI. Kamau (2019) looked at how governance quality and political stability affect FDI in East Africa.

From his research, it is evident that better governance significantly attracts FDI. The study excludes the environmental and social components of ESG and it also failed to perform a comparative analysis that would measure the relative importance of governance against all other ESG factors.

Having looked at their valuable contributions, it is quite evident that there remains room for further exploration, specifically the combined ESG factors and their impact on FDI. This research aims to dig deeper into this concept, looking to compare the relative importance of different ESG factors in attracting FDI. Understanding which ESG factors are most critical to investors could provide valuable insights for policymakers for better implementation. By achieving this, it seeks to provide a better understanding of market patterns that investors can use during their evaluation of foreign investments. Many previous studies are country-specific and do not offer sufficient regional analysis of East Africa as a whole. A broader perspective could identify regional trends and common challenges or opportunities related to ESG and FDI. This can further be used to develop a regional index, a tool for investors within the East African markets. Lastly, limited research has been conducted from the perspective of foreign investors, particularly in terms of how they weigh ESG factors against traditional economic factors when making investment decisions. Thus, while building on the groundwork laid by previous studies, this study seeks to fill these critical gaps by illustrating the relationship between ESG factors and FDI.

3. Methodology

The "push vs. pull" framework highlights key factors in ESG that are possible drivers of FDI. Country-specific (pull) factors demonstrate strong correlation between FDI to emerging markets. Koepke (2015) , posits that domestic output growth as well as domestic country risk indicators have shown a significant impact on FDI for emerging and developing countries. My hypothesis borrows from the above and it posits that ESG factors have a significant impact on FDI inflows. I shall test the hypothesis through panel regression models. "Compared to purely cross-sectional data, panels are attractive since they often contain far more information than single cross-sections and thus allow for an increased precision in estimation." - Daniel Hoechle.
As opposed to pure cross-sectional data, panel data is structured in wide or long formats with multiple dimensions such that we have cross-sectional (i = 1, …, N) and a time series (t = 1, …, T) and cross-sectional (i = 1, …, N). The equation for the model then becomes:

$$y_{it} = \alpha + \beta x_{it} + e_{it} \qquad (I)$$

Where:

- y_{it} being the dependent variable
- α represents the intercept
- β is the vector of estimates
- x_{it} are the explanatory variables
- e_{it} are the error terms

With the panel regression approach, I shall be able to estimate the change in correlation between variables over time, a solution to the challenge of having little data to analyze over the years selected. It would otherwise be difficult to run the analysis for individual countries with very few data points. The first advantage is that it enables examination of how variable relationships change with time. Having both time series and cross-sectional data as one provides a higher degree of freedom, compensating for the loss of power due to minimal data points for analysis.

For the country-fixed model, the error term e_{it} from equation (I) shall be decomposed into two parts resulting to the following equation:

$$y_{it} = \alpha + \beta x_{it} + \mu_{it} + v_{it} \qquad (II)$$

Here, μ_{it} is the country-fixed effect accounting for all the variables impacting y_{it} cross-sectionally without a time-variant. v_{it} on the other hand, accounts for any disturbance that remains unexplained by the model. The Least Squares Dummy Variable (LSVD) is hence an appropriate approach to estimate this model, which culminates in the following structure:

$$y_{it} = \alpha + \beta x_{it} + \mu_1 D1_i + \mu_2 D2_i + \ldots + \mu_N DN_i + v_{it} \qquad (III)$$

For all available observations on each ith variable (country), the dummy variable $Di_i (i = 1, \ldots, N)$ equals 1, and 0 otherwise. This therefore means that the regression model returns the same vector of β, albeit with a different intercept value. These intercepts are country-specific and they reflect the ESG to FDI correlation that's uniquely tied to a country.

The first step in the analysis will conduct a pooled regression model that does not distinguish data for the different countries. The final regression model shall incorporate country-specific (fixed) effects to ensure the inclusion of any unique unidentified explanatory variables tied to each country. The next section of this paper focuses on the regression model in detail.

Justification for Methodology

The panel regression models were chosen for this study due to their ability to capture both country-specific effects and temporal dynamics. Compared to purely cross-sectional data, panel data often contain more information, allowing for increased precision in estimation. Panel regression models can effectively handle the complexity arising from having multiple countries and years in the data, providing a robust analysis of the relationship between ESG factors and FDI inflows.

Variable Selection

The variables selected for this study are chosen based on their relevance to ESG factors and their potential impact on FDI inflows. Each of them represents a specific aspect of environmental, social, or governance factors, which are crucial for understanding the relationship between ESG and FDI. Below is a detailed explanation of the relevance of each variable:

1. Renewable Energy Consumption (% of total final energy consumption)

This variable is a measure of a country's commitment to sustainable energy practices. Higher renewable energy consumption reflects efforts to reduce carbon footprints and dependence on fossil fuels, which makes the country more attractive to environmentally conscious investors. It also signifies long-term sustainability and a proactive approach to environmental conservation.

2. CO2 Emissions (metric tons per capita)

CO2 emissions are a direct measure of environmental pollution. Lower emissions indicate better environmental practices, which can attract foreign investors who prioritize sustainability. Reducing emissions is often associated with modern and efficient technologies, as well as regulations that can positively impact investment decisions.

3. Control of Corruption (Estimate value)

This is a governance variable that measures the perceived level of corruption in a country. Higher control of corruption suggests a transparent and fair business environment. This measure is key for attracting foreign investors. On the contrary, corruption can increase the risk and cost of doing business, negatively affecting investment.

4. Government Expenditure on Education (% of GDP)

Investment in education represents a country's focus on building human capital. Higher government expenditure on education indicates a well-educated workforce, which is a significant factor for foreign investors seeking skilled labor. It also reflects social stability and long-term economic growth potential.

5. Unemployment Rate (total % of labor force)

The unemployment rate is a critical social factor that affects the overall economic health of a country. Lower unemployment rates suggest a robust labor market, economic stability, and higher consumer spending power. These factors make a country more attractive for investment. On the other hand, high unemployment rates can indicate economic distress and social instability.

6. GDP Growth (annual %)

GDP growth is a primary indicator of economic performance. Higher GDP growth rates signal a growing economy, which can attract FDI by promising better returns on investment. It indicates economic stability and potential for future growth, which are key considerations for foreign investors.

7. Government Effectiveness (Estimate value)

This variable measures the quality of public services, policy formulation, and implementation. Higher government effectiveness reflects a stable and efficient administration, which is essential for creating a conducive business environment. Effective governance reduces risks and uncertainties for investors.

8. Political Stability and Absence of Violence/Terrorism (Estimate value)

Political stability is a crucial factor for investment decisions. Countries with high political stability and low levels of violence or terrorism are more likely to attract FDI, as they offer a safer and more predictable business environment. Political unrest can deter investment due to increased risks and potential disruptions.

9. Agriculture, Forestry, and Fishing (value added % of GDP)

This environmental variable highlights the importance of the primary sector in the economy. A significant contribution from agriculture, forestry, and fishing indicates the potential for sustainable practices and resource management, which can be attractive to investors focused on environmental sustainability.

The regression model

Table 1. Overview of regression variables

Variable		**Notation**	**Description**
Dependent	y	FDI	Foreign direct investment, net inflows (% of GDP)
Explanatory	X1	GDP	Gross Domestic Product growth (annual %)
Explanatory (ESG)	X2	AGRF	Agriculture, forestry, and fishing, value added (% of GDP)
	X3	CO2	CO2 emissions (metric tons per capita)
	X4	CoC	Control of Corruption: Estimate
	X5	EDXPD	Government expenditure on education, total (% of GDP)
	X6	GE	Government Effectiveness: Estimate
	X7	POLS	Political Stability and Absence of Violence/Terrorism: Estimate
	X8	RE	Renewable energy consumption (% of total final energy consumption)
	X9	UEM	Unemployment, total (% of total labor force) (modeled ILO estimate)

Pooled Ordinary Least Squares (OLS) Regression

As mentioned earlier, the first step of panel data analysis will fit a pooled regression model combining all observations across different countries and years, handled as one dataset. This will provide an overall view of the relationships between ESG factors and FDI, with the dependent variable being FDI inflows as a percentage of GDP. The equation then becomes:

The pooled OLS regression combines all the data points across different countries and years, treating it as a single dataset. The mathematical representation of the pooled OLS model is:

$$FDI_{it} = \alpha + \beta AGRF_{it} + \beta CO2_{it} + \beta CoC_{it} + \beta EDXPD_{it} + \beta GDP_{it} + \beta GE_{it} + \beta POLS_{it} + \beta RE_{it} + \beta UEM_{it} + \epsilon_{it}$$

Where:

- $_i$ is the country
- $_t$ represents the year
- α is the intercept
- $\beta_1, \beta_2, \ldots, \beta_9$ are the coefficients for the respective independent variables
- ϵ_{it} is the error term

The pooled OLS regression shall be performed using the *statsmodels* library in Python, and the summary of results obtained.

Fixed Effects Regression with Heteroscedasticity-Robust Standard Errors

The fixed effects regression model covers for unobserved heterogeneity across selected countries by including country-specific effects. Heteroscedasticity-robust standard errors shall be used to adjust the standard errors of the model coefficients. This accounts for possible heteroscedasticity. Heteroscedasticity causes bias in ordinary least squares (OLS) standard errors, resulting in incorrect conclusions about the variable significance. Robust standard errors therefore adjust for this variance instability and provide more reliable standard error estimates, ensuring that hypothesis tests remain valid despite the presence of heteroscedasticity.

With this approach, we introduce control for time-invariant characteristics of each country. The fixed model can be represented as:

$$FDI_{it} = \alpha + \beta AGRF_{it} + \beta CO2_{it} + \beta CoC_{it} + \beta EDXPD_{it} + \beta GDP_{it} + \beta GE_{it} + \beta POLS_{it} + \beta RE_{it} + \beta UEM_{it} + \mu_{it} + \epsilon_{it}$$

Where:

- μ_{it} represents the country-specific fixed effects

This can be achieved using the *cov_type='HC1'* argument in the statsmodels library in Python. The fixed effects regression model shall first include dummy variables for each country, and then fit the OLS regression with these dummies included.

Data Acquisition

The analysis shall be conducted using open-source data on ESG factors for several East African countries. Specifically, I shall look at data for 6 emerging and developing countries in the region (East African Development Bank, 2023) as presented in Appendix 1. The sample focuses on annual data collected over a period of 5 years (2017-2022) with a key interest in countries with readily available data for the chosen ESG variables.

For the analysis, FDI inflows are the dependent variable. This is collected on a net basis for each country i.e., I consider liabilities void of all assets traded between foreign investors and host countries. Negative FDI values imply reverse investment or lack thereof (the World Bank, 2024). Independent variables used comprise select factors within each of the three aspects of ESG. The variables under the Environmental factors are Renewable energy consumption (% of total final energy consumption), CO2 emissions (metric tons per capita), and Agriculture, forestry, and fishing, value added (% of GDP). Social factors selected are Government expenditure on education, total (% of GDP) and Unemployment, total (% of total labor force) (modeled ILO estimate). Lastly, the Governance factors are Control of Corruption (Estimate value), Government Effectiveness and Political Stability, and Absence of Violence/Terrorism. Data for all the factors mentioned above are fetched from the ESG Database as provided in the World Bank open databank repository.

Existence of outliers in some explanatory variables could lead to biased results in the regression parameter estimates. It is therefore important to identify and appropriately handle such cases to ensure the robustness and reliability of the models. For this, I shall use the interquartile range method to identify and remove outliers in the data.

I shall use Python to conduct the above analysis, the results of the same shall be presented in the appendix.

Possible Obstacles and Impediments

In this section, I highlight some possible challenges that may arise during implementation of the research:

Data Availability and Quality

The availability of high-quality, comprehensive, and up-to-date data on ESG factors and FDI inflows for East African countries can be a significant challenge. Incomplete or inconsistent data can affect the robustness of the analysis. To address this issue, data will be sourced from reliable and reputable sources such as the World Bank's open data

repository. Additionally, techniques like interpolation and data imputation may be employed to handle any missing data points.

Heterogeneity of Data

ESG factors vary by nature, this is for both quantitative and qualitative measures. This variation can present challenges in standardizing the data for analysis. Employing composite indices or data normalization techniques can help standardize disparate data types. This will then allow for a more coherent analysis.

Model Specification and Assumptions

Choosing the appropriate model specification and ensuring that all assumptions (e.g., no multicollinearity, homoscedasticity) are met can be complex and critical for the validity of the results. As part of model validation, I will conduct testing for model assumptions and explore alternative model specifications that can enhance the robustness of the findings.

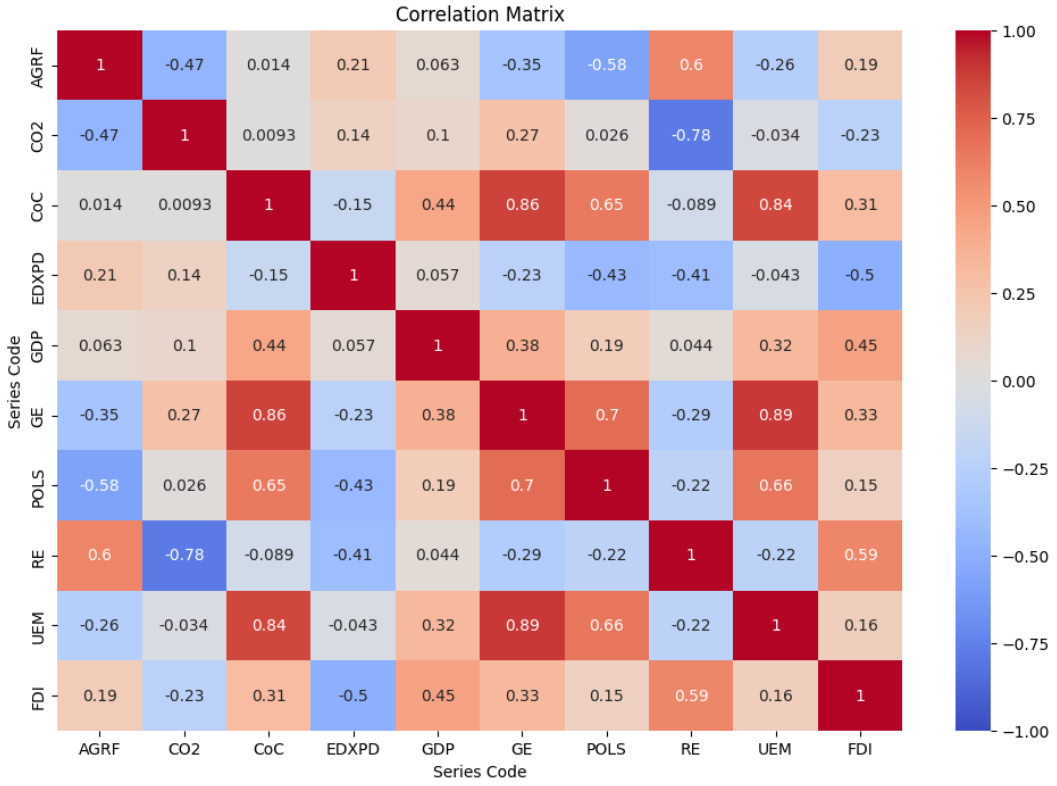
Policy and Institutional Barriers

Given the analysis is done across the East African region, variations in policy and institutional frameworks across different countries can impact the generalizability of the findings. Incorporating country-specific fixed effects in the regression models can help account for these variations and provide better insights.

4. Empirical Results and Discussion

Correlation Analysis

Figure 1. Overview of variable correlation



The results above depict some key relationships between ESG factors and Foreign Direct Investment (FDI). A strong positive correlation between renewable energy usage (RE) and FDI (0.59) implies that countries investing in renewable energy tend to attract more FDI.

Government effectiveness (GE) and control of corruption (CoC) show moderate positive correlations with FDI (0.33 and 0.31) adding importance to good governance and corruption control in attracting FDI.

Education expenditure (EDXPD) has a moderate negative correlation with FDI (-0.5), which means that higher spending on education might be associated with lower immediate FDI inflows.

On the other hand, CO2 emissions show a weak negative correlation with FDI (-0.23) which tells us that environmental concerns might slightly deter foreign investments. Other factors such as agricultural production (AGRF), political stability (POLS), and unemployment (UEM) show weak correlations with FDI.

Regression Analysis

Table 2. Summary of Pooled OLS Regression Results

Pooled OLS Regression Results:

OLS Regression Results

Dep. Variable:	FDI	R-squared:	0.795
Model:	OLS	Adj. R-squared:	0.724
Method:	Least Squares	F-statistic:	11.21
Date:	Sun, 07 Jul 2024	Prob (F-statistic):	6.32e-07
Time:	07:53:02	Log-Likelihood:	-37.977
No. Observations:	36	AIC:	95.95
Df Residuals:	26	BIC:	111.8
Df Model:	9		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-11.4222	7.997	-1.428	0.165	-27.860	5.016
AGRF	-0.0540	0.105	-0.513	0.612	-0.270	0.162
CO2	1.7941	5.305	0.338	0.738	-9.111	12.699
CoC	-0.2978	0.963	-0.309	0.760	-2.277	1.682
EDXPD	0.1373	0.281	0.489	0.629	-0.439	0.714
GDP	0.1221	0.062	1.981	0.058	-0.005	0.249
GE	3.5502	1.575	2.254	0.033	0.313	6.788
POLS	-0.0839	0.707	-0.119	0.907	-1.538	1.370
RE	0.1973	0.072	2.728	0.011	0.049	0.346
UEM	-0.2147	0.174	-1.230	0.230	-0.573	0.144

Omnibus:	11.686	Durbin-Watson:	1.414
Prob(Omnibus):	0.003	Jarque-Bera (JB):	13.282
Skew:	0.946	Prob(JB):	0.00131
Kurtosis:	5.296	Cond. No.	6.14e+03

The Pooled OLS regression combines all ESG and FDI variables across different countries and years, handling them as a single dataset. The model identifies the overall relationship between ESG factors and FDI inflows without accounting for country-specific effects.

The pooled OLS regression model explains 79.5% of the variance in FDI (R-squared = 0.795). Significant variables include *Government Effectiveness (GE)* and *Renewable Energy (RE)*, both positively impacting FDI. This model may however suffer from omitted variable bias as it does not account for unobserved heterogeneity across countries. To cater for this possibility, the fixed-effects model is fitted.

Table 3. Summary of Fixed Effects Regression Results

Fixed Effects Regression Results with Country-Specific Effects:
OLS Regression Results

Dep. Variable:	FDI	R-squared:	0.853
Model:	OLS	Adj. R-squared:	0.755
Method:	Least Squares	F-statistic:	33.15
Date:	Sun, 07 Jul 2024	Prob (F-statistic):	3.00e-11
Time:	08:11:37	Log-Likelihood:	-32.017
No. Observations:	36	AIC:	94.03
Df Residuals:	21	BIC:	117.8
Df Model:	14		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	7.7278	16.543	0.467	0.640	-24.697	40.152
Q("Country Name_Burundi")[T.True]	-2.2589	4.957	-0.456	0.649	-11.974	7.457
Q("Country Name_Ethiopia")[T.True]	1.3123	2.800	0.469	0.639	-4.176	6.800
Q("Country Name_Kenya")[T.True]	-2.2391	2.950	-0.759	0.448	-8.022	3.544
Q("Country Name_Rwanda")[T.True]	6.3550	2.630	2.417	0.016	1.201	11.509
Q("Country Name_Tanzania")[T.True]	0.4776	1.893	0.252	0.801	-3.232	4.188
Q("Country Name_Uganda")[T.True]	4.0809	6.986	0.584	0.559	-9.611	17.773
AGRF	-0.0116	0.165	-0.070	0.944	-0.334	0.311
CO2	0.6656	11.909	0.056	0.955	-22.676	24.007
CoC	-0.0615	4.633	-0.013	0.989	-9.142	9.019
EDXPD	0.5219	0.536	0.974	0.330	-0.528	1.572
GDP	0.0713	0.070	1.024	0.306	-0.065	0.208
GE	1.6032	1.370	1.171	0.242	-1.081	4.288
POLS	-1.2053	0.874	-1.380	0.168	-2.917	0.507
RE	-0.0811	0.214	-0.379	0.705	-0.501	0.338
UEM	-0.5683	0.447	-1.271	0.204	-1.445	0.308

Omnibus:	15.179	Durbin-Watson:	1.485
Prob(Omnibus):	0.001	Jarque-Bera (JB):	18.918
Skew:	1.203	Prob(JB):	7.80e-05
Kurtosis:	5.612	Cond. No.	1.52e+19

The fixed effects regression model helps to account for unobserved heterogeneity across countries by including country-specific effects. With this approach, we control for time-variant characteristics of each country that may influence FDI inflows. In the presence of heteroscedasticity, consistent standard errors can be achieved by using heteroscedasticity-robust standard errors to ensure the reliability of the output.

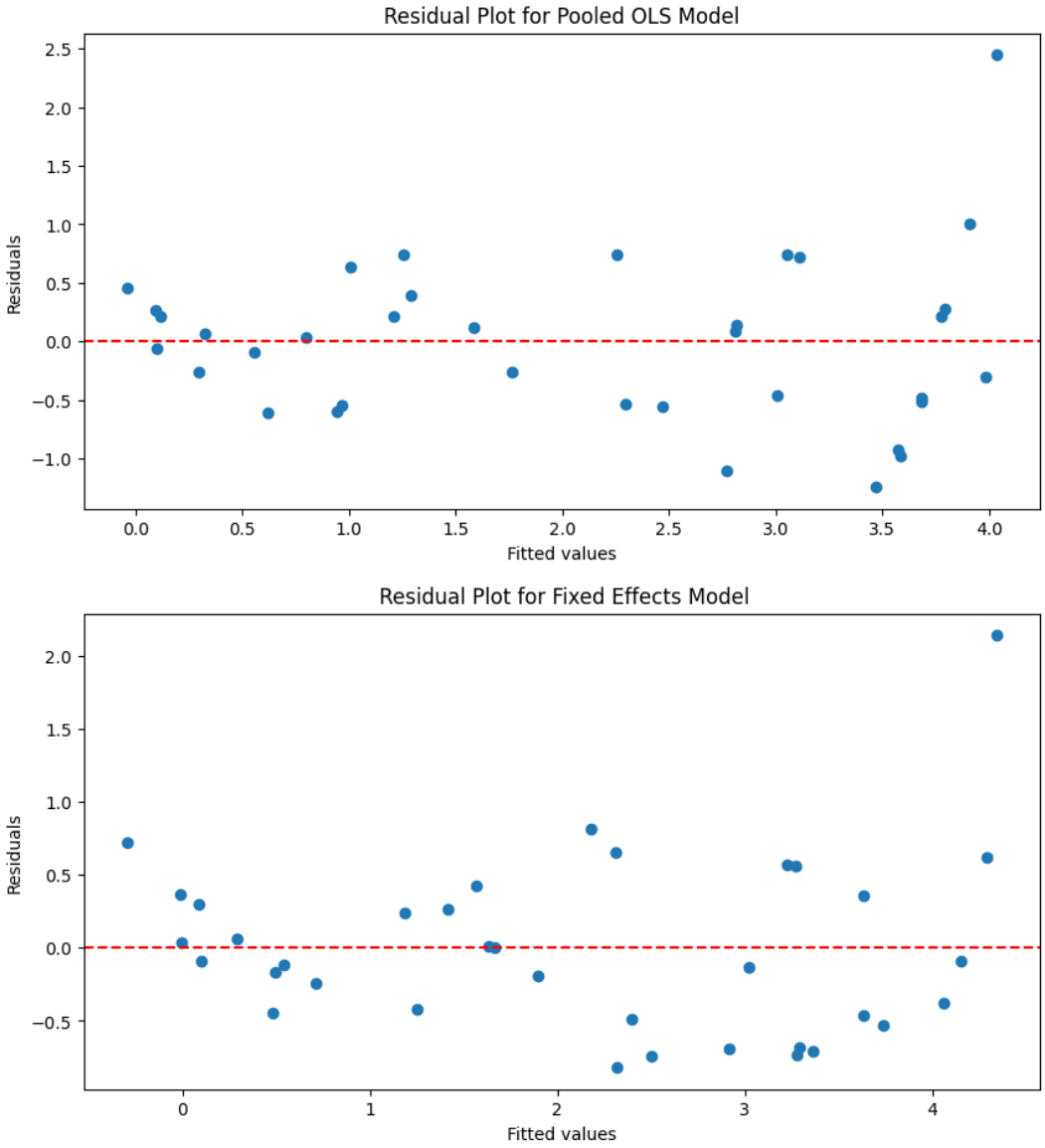
This model includes dummy variables for each country to capture country-specific effects. This significantly improves the explanatory power to 85.3% (R-squared = 0.853). This model confirms the significance of GE and RE having their coefficients remain positive and statistically significant.

The model also shows significant variation in FDI inflows across countries. It also addresses heteroscedasticity by using robust standard errors. From the output, we see that while general factors like GE and RE are crucial, country-specific characteristics also play a significant role in attracting FDI.

Model Assumptions and Predictability

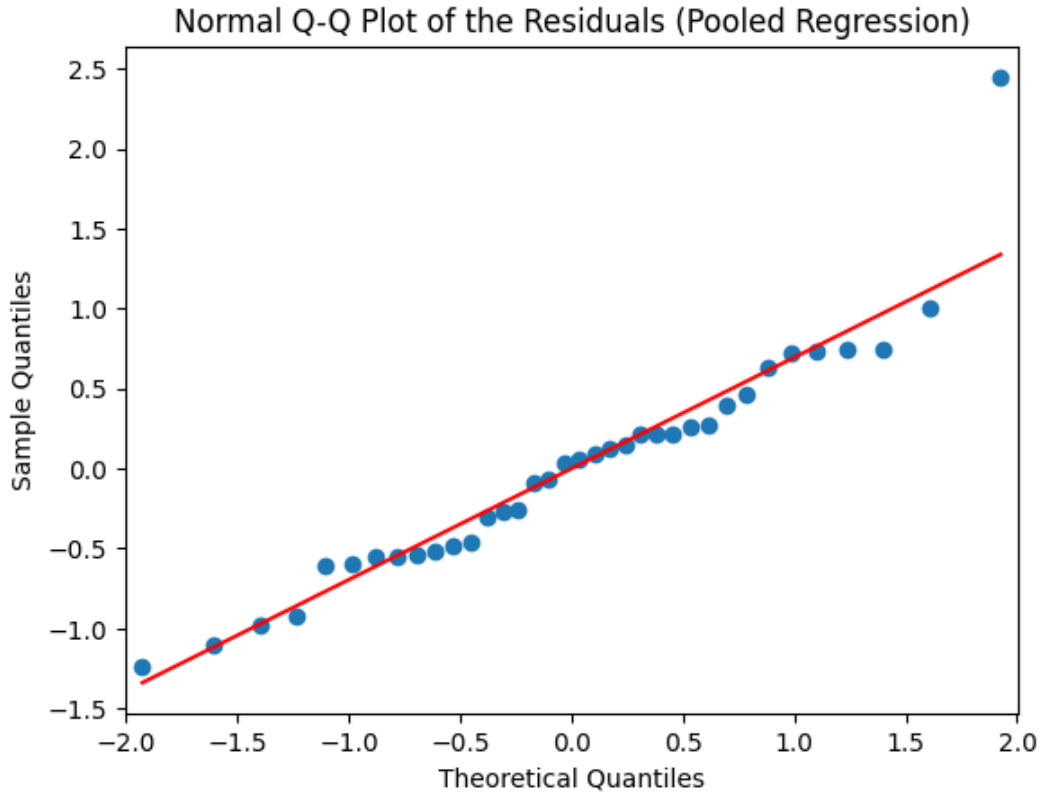
This section looks into the assumptions, the quality of the fit, and the predictability of the model based on the pooled OLS and fixed effects regression models.

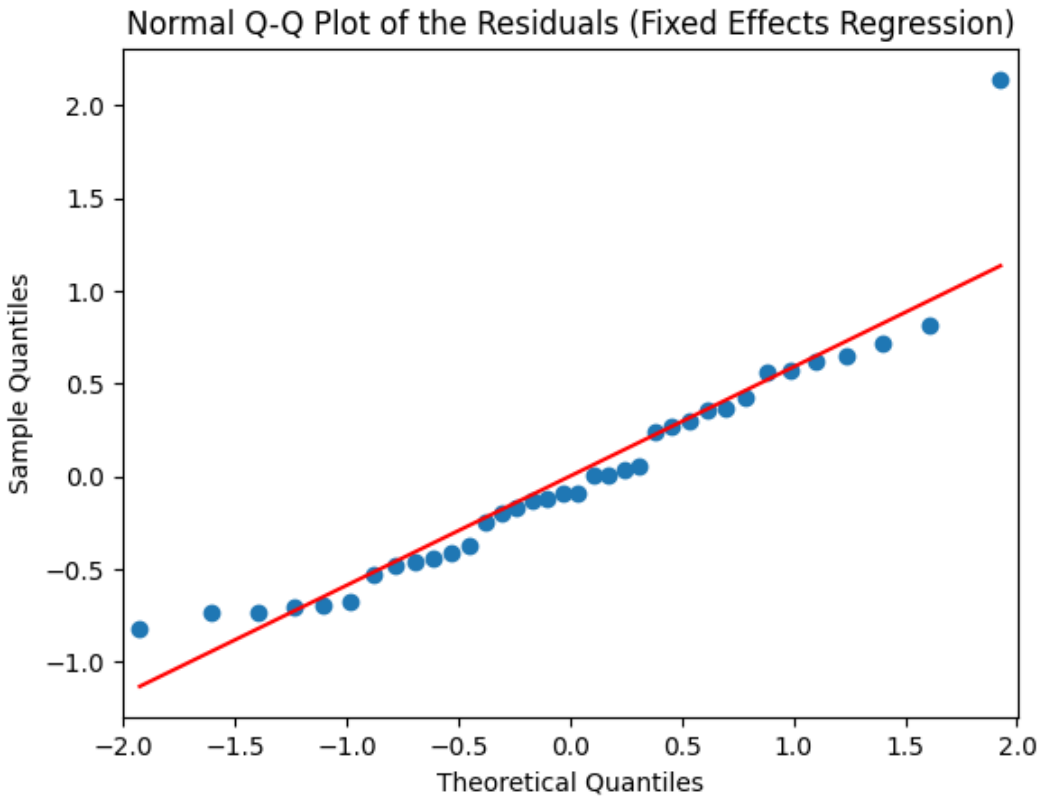
Figure 2: Residual Plots



From the plots above, neither plot shows clear evidence of problematic patterns such as a funnel shape (which would indicate heteroscedasticity) or a curve (indicating non-linearity), but the presence of outliers suggests that further robustness checks or transformations might be required. The Fixed Effects model appears to control better for unobserved heterogeneity. This is evident from the fewer and less extreme outliers compared to the Pooled OLS model. This might suggest a more accurate model fit when accounting for individual-specific variability.

Figure 3: Q-Q Plots





For the pooled regression model, the residuals largely follow the theoretical line. This implies that they are approximately normally distributed across most of their range. However, a notable deviation at the upper end of the distribution indicates the presence of outliers or extreme values. These outliers appear to be pulling away from the normal line, suggesting that the upper tail is heavier than expected under normality.

Similar to the pooled regression, the residuals from the fixed effects model mostly align with the theoretical line, indicating that the residuals are close to normal. Like the pooled model, there is a significant deviation in the upper tail, which points to the presence of outliers or extreme values that do not conform to a normal distribution.

Table 4: Comparison of goodness of fit and model complexity

Model	AIC	BIC
Pooled OLS Regression	95.9546	111.7894
Fixed Effects Regression	94.8344	177.7872

The comparison of AIC and BIC values between the Fixed Effects Model and the Pooled OLS Model provides insights into their relative performance in terms of fit and complexity. The Fixed Effects Model exhibits a slightly lower AIC (94.83) compared to the Pooled OLS Model (95.95), suggesting that the Fixed Effects Model fits the data slightly better when the number of parameters is taken into account. This is indicative of the Fixed Effects Model capturing more of the variance in the dependent variable due to incorporating individual-specific variations or unobserved heterogeneity, which the Pooled OLS Model does not account for.

However, the BIC, which includes a larger penalty for the number of parameters, shows a significantly higher value for the Fixed Effects Model (177.79) compared to the Pooled OLS Model (111.79). This indicates that the increase in model complexity with the Fixed Effects Model may not be justified given the data, as per the BIC's criteria. BIC's higher value suggests caution against overfitting, particularly in scenarios where model simplicity is important.

5. Conclusion

The analysis conducted in this study aimed to assess the impact of Environmental, Social, and Governance (ESG) factors on Foreign Direct Investment (FDI) across various countries within the East African Region, using both Fixed Effects and Pooled OLS models. The findings indicate that environmental factors, such as CO2 emissions, tend to have a weak to moderate negative correlation with FDI. This simply suggests that higher emissions could deter investment due to growing global concerns and regulatory pressures on sustainability. It is therefore recommended that governments should focus on implementing policies to control CO2 emissions to promote Foreign Direct Investment.

Social factors like education expenditure showed mixed effects. While in some cases these factors proved to be beneficial in the long-term, they do not necessarily attract immediate investment. On the other hand, governance factors such as political stability and government effectiveness consistently exhibited a positive correlation with FDI,

emphasizing the importance of stable and efficient governance structures in attracting investment. Additionally, governments are encouraged to develop policies that promote economic growth through FDI. This could be achieved by reducing taxes on imports that are required for production, reducing export duties, and also putting more focus on infrastructure development initiatives. Examples of such infrastructure are telecommunication, access to electricity, road and rail networks among others.

While the Fixed Effects Model provided a deeper understanding of individual country dynamics and showed a better fit as indicated by a lower AIC, its higher BIC suggested potential overfitting. Contrary to this, the Pooled OLS Model offered a broader perspective that can be useful for determining policy recommendations. The statistical significance of these factors provided different highlights on specific areas for policymakers to focus on improving their country's attractiveness to investors.

Future research could explore the impact of specific ESG initiatives in more detail and validate the models across different contexts to enhance the generalizability of these findings. This study underlines the critical role of robust governance in attracting foreign investment and suggests a balanced approach in policy formulation that considers economic returns alongside long-term social and environmental benefits.

Other Possible Real-Life Applications of the Research

Over and above the primary objective to understand the impact of ESG factors on FDI inflows in East African countries, this research has several other potential real-life applications. These are but not limited to:

1. *Policy Formulation*

   Policymakers can use the insights from this research to design and implement policies that enhance ESG standards, thereby attracting higher levels of FDI. Improved ESG practices can lead to sustainable economic growth and development.

2. *Investment Decision-Making*

   Investors can leverage the findings to make informed decisions regarding investment destinations. Understanding which ESG factors are most significant to FDI inflows can help investors allocate resources more effectively and align their investments with sustainability goals.

3. *Corporate Strategy and Governance*

   Corporations can utilize the research to refine their ESG strategies and practices. Companies operating in or targeting East African markets can improve their ESG performance to enhance their attractiveness to foreign investors.

4. *Academic and Educational Use*

   The research can serve as a valuable resource for academic studies and educational purposes. It can contribute to the body of knowledge on ESG and

FDI, offering case studies and empirical evidence for further research and learning.

5. *Development Programs and Aid*

International development agencies and non-governmental organizations (NGOs) can use the insights to design programs that promote sustainable development. By understanding the role of ESG factors, these organizations can better support initiatives that drive both economic and social progress.

6. *Benchmarking and ESG Index Development*

The creation of a comprehensive ESG Index for East African markets can serve as a benchmark for comparing ESG performance across countries. This index can guide investors, policymakers, and other stakeholders in evaluating and improving ESG standards.

By addressing potential challenges encountered during implementation, this research can be posited as a comprehensive and impactful study with broad relevance and utility for the East African markets.

Appendixes

Appendix 1. List of countries used.

Country	Region
Burundi	East Africa
Ethiopia	
Kenya	
Rwanda	
Tanzania	
Uganda	

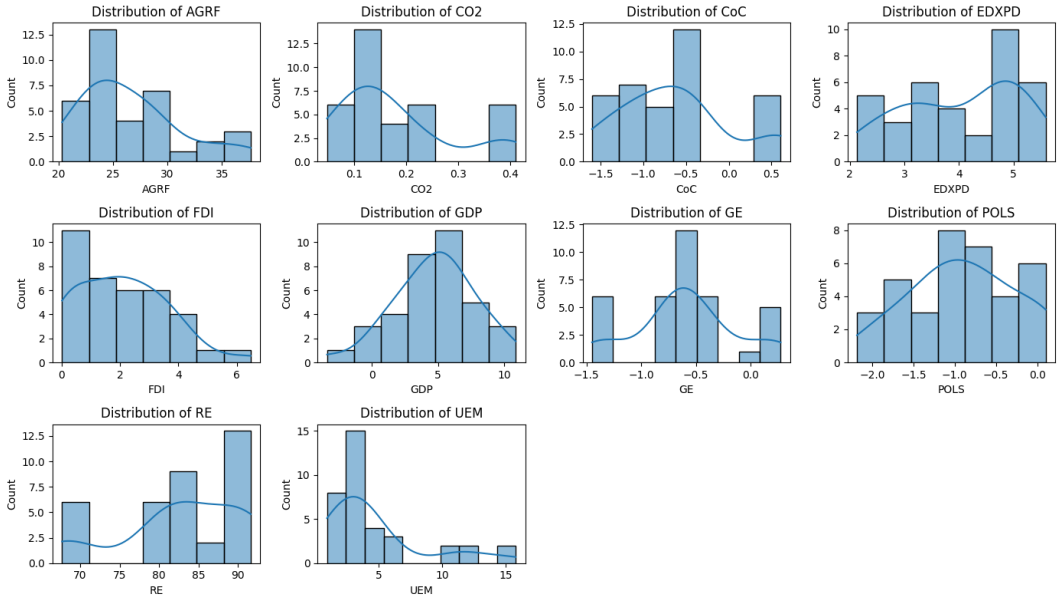
Appendix 2. Descriptive Statistics

Summary Statistics:

Series Code	AGRF	CO2	CoC	EDXPD	FDI	GDP
count	36.000000	36.000000	36.000000	36.000000	36.000000	36.000000
mean	26.591511	0.176453	-0.622591	4.034847	2.082628	4.753152
std	4.597123	0.109190	0.652798	1.051532	1.557155	3.040865
min	20.314910	0.047429	-1.599391	2.130520	0.011620	-3.373694
25%	23.535185	0.104970	-1.055182	3.204430	0.457007	3.062827
50%	25.070938	0.144051	-0.659202	4.260755	1.837194	4.980397
75%	28.683535	0.222250	-0.405915	4.868666	3.037216	6.519628
max	37.642228	0.410582	0.607265	5.590650	6.480676	10.857967

Series Code	GE	POLS	RE	UEM
count	36.000000	36.000000	36.000000	36.000000
mean	-0.593589	-0.904148	82.794444	4.665333
std	0.467256	0.650295	7.746251	3.960172
min	-1.449708	-2.183713	67.700000	0.906000
25%	-0.749190	-1.281865	80.275000	2.563500
50%	-0.639647	-0.940415	83.050000	3.398500
75%	-0.396814	-0.457724	90.500000	5.164500
max	0.274038	0.101643	91.600000	15.790000

Appendix 3. Data Distribution



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Authors

Author's picture should be in grayscale.

Picture size should be absolute 3.18cm in height and absolute 2.65cm in width

Mwange Carlvin Jerry